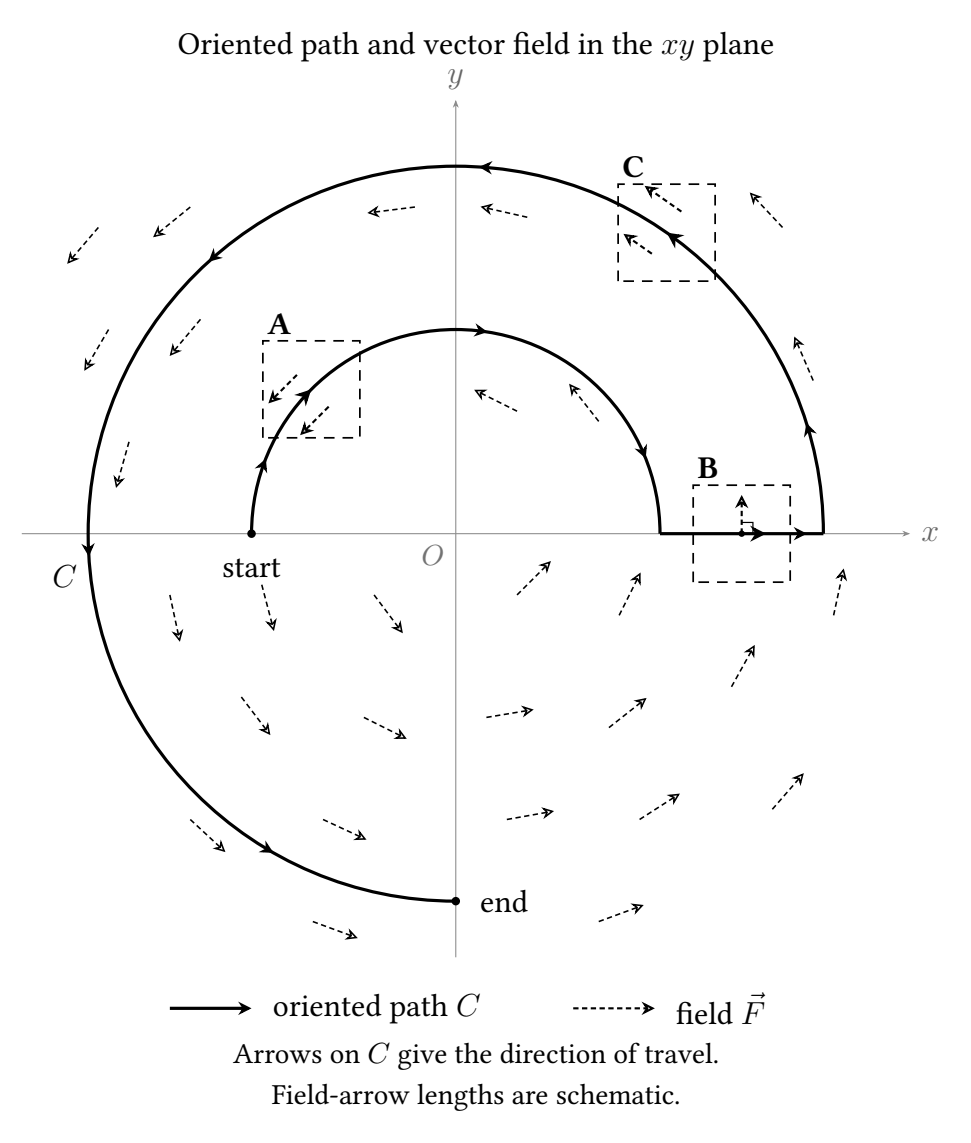

Oriented path and vector field in the xy plane
y
x
O
C
start
end
A
B
C
oriented path C
field $\vec{F}$
Arrows on C give the direction of travel.
Field-arrow lengths are schematic.